



Effects of climate change on insect pests dynamics

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Climate change

Climate change is the term used to describe a gradual increase in the average temperature of the Earth's atmosphere and its oceans, a change that is believed to be changing the Earth's climate forever. Global temperature has been steadily rising since 1900 with an increase of about 1°C since then. The greatest increase has been in northwestern North America, but India's temperature has increased between 0.2°C and 1°C.

Moreover, the rate of global warming is increasing; temperature increased twice as fast during the last 50 years as it did in the last 100 years. The mean temperature in India is projected to increase up to 1.7°C in *kharif* (July to October) and up to 3.2° C during *rabi* (November to March) season, while the mean rainfall is expected to increase by 10 percent by 2070 (Gupta, 2011).

Impacts of climate change on insect pests

Insects are cold-blooded organisms - the temperature of their bodies is approximately the same as that of the environment. Therefore, temperature is probably the single most important environmental factor influencing insect behaviour, distribution, development, survival, and reproduction.

Anthropogenic CO₂ is almost twice more important for temperature increase than other long-lived greenhouse gases combined. Although increased CO₂ should not directly deleteriously affect insects, the temperature increases driven by the increase in anthropogenic CO₂ already affect insects in profound ways including their distribution, nutrition, phenology and role as disease vectors.

Effects of elevated CO₂ on insect pests

In general, host plants grown under elevated CO₂ are less nutritious to insect herbivores, which can affect their behaviour and performance. Phenotypic host-plant changes typically make leaf material eaten by insects less nutritious. As a consequence, insects have a more difficult time converting the food they eat into biomass. In order to mitigate the effects of less nutritious food, insect herbivores often consume more.

Insect herbivore performance is positively correlated with leaf nitrogen concentrations. Zvereva and Kozlov (2010) reported that the leaf nitrogen content decreased for mustard and collard grown under elevated CO₂. Leaf chewing insect herbivore performance is positively correlated with leaf water content. Decrease in leaf water contents was observed under elevated CO₂ for both mustard and collard.

Plants can also defend themselves mechanically, either by having tough leaves or by structures such as leaf trichomes. Levels of mechanical defense are negatively correlated with herbivore performance. Elevated CO₂ increased trichome densities on radish. Several studies, mostly considering leaf toughness, leaf thickness, and specific leaf weight, have also observed increases in mechanical defense due to elevated CO₂. Hamilton *et al.* (2005) reported higher percent leaf damage or consumption due to cabbage white butterfly fed either mustard or collard grown under elevated CO₂. Similar results have been obtained in leaf miners on a variety of woody species. Zvereva and Kozlov (2010) detected a significant negative effect of elevated CO₂ on insect herbivore performance. They observed that overall herbivore communities were lower on plants grown under elevated CO₂ vs. ambient CO₂. This is likely in part due to higher mortality rates due to both parasitoids and other natural enemies. Natural enemies are thought to have better success under elevated CO₂ because their prey are more apparent. Insects typically take longer to develop, making them more apparent in time to natural enemies. Higher consumption rates also cause increased leaf damage and increased frass production, both cues to natural enemies.

Hamilton *et al.* (2005) measured levels of herbivory in soybean grown in ambient air and air enriched with CO₂ or O₃ using free air gas concentration enrichment (FACE). Under open-air conditions and exposure to the full insect community, elevated CO₂ increased the susceptibility of soybeans to herbivory early in the season, whereas exposure to elevated O₃ seemed to have no effect. In the region of the canopy exposed to high levels of herbivory, the percentage of leaf area removed increased from 5 to more than 11% at elevated CO₂. They found no evidence for compensatory feeding at elevated CO₂ where leaf nitrogen content and C:N ratio were unaltered in plants experiencing increased herbivory. However, levels of leaf sugars were increased by 31% at elevated CO₂ and coincided with a significant increase in the density of the invasive species *Popillia japonica* (Japanese beetle). In two-choice feeding trials, Japanese beetles and Mexican bean beetles (*Epilachna varivestis*) preferred foliage grown at elevated CO₂ to foliage grown at ambient CO₂. Hence, the increased level of sugar in leaves grown at elevated CO₂ may act as a phagostimulant for the Japanese beetle.

Effects of elevated temperature on insect pests

Many of the effects of increased temperature on insect performance have to do with the direct effects of temperature on insects. Because insects are exothermic, they tend to be more active under warmer conditions. A typical effect of elevated temperature is therefore to increase consumption rates and therefore decrease the time to pupation, making them less apparent to natural enemies and in some cases increasing the potential number of generations per season. It has been estimated that with a 2° C temperature increase insects might experience one to five additional life cycles per season (Yamamura and Kiritani, 1998). Elevated temperatures increase gypsy moth performance, both decreasing its development time and increasing its survival rate (Williams *et al.*, 2003). However the survival rate of another member of its genus, the nun moth, is very different under increased temperatures. If gypsy moths react more favorably to future environments than competitors, they may become more prone to outbreak. Elevated temperatures (on the scale of expected global warming) can also have direct effects on plant phenotypes, but not typically to the extent that elevated CO₂ has, and those factors affected (like total nonstructural carbohydrates, starches, and sugars) don't typically affect insect herbivores as much as host-plant characteristics affected by elevated CO₂.

Temperature may change gender ratios of some pest species such as thrips (Lewis, 1997) potentially affecting reproduction rates. Insects that spend important parts of their life histories in the soil may be more gradually affected by temperature changes than those that are above ground simply because soil provides an insulating medium that will tend to buffer temperature changes more than the air (Bale *et al.*, 2002). Lower winter mortality of insects due to warmer winter temperatures could be important in increasing insect populations (Harrington *et al.*, 2001). Insect species diversity per area tends to decrease with higher latitude and altitude (Andrew and Hughes, 2005), meaning that rising temperatures could result in more insect species attacking more hosts in temperate climates (Bale *et al.*, 2002).

Effect of changes in rainfall pattern on insect pests

Early and timely planting become more uncertain under climate change. During the 2009 rainy season, delay in onset of monsoons by 45 days resulted in delayed plantings of pigeonpea that are prone to damage by *Helicoverpa armigera* and caused heavy damage (Sharma, 2010). As with temperature, precipitation changes can impact insect pest predators, parasites, and diseases resulting in a complex dynamic. Fungal pathogens of insects are favored by high humidity and their incidence would be increased by climate changes that lengthen periods of high humidity and reduced by those that result in drier conditions. Some insects are sensitive to precipitation and are killed or removed from crops by heavy rains, this consideration is important when choosing management options for onion thrips (Reiners and Petzoldt, 2005).

Conclusion

Species life history (evolutionary) adaptations may obscure our ability to detect species response to climate change - accordingly, species respond differently to changes in thermal environments. There are many interactions and it is extremely difficult to predict the impact of climate change on insect pests in the future, but we may expect an increase of certain primary pests as well as secondary pests and invasive species. The best economic strategy for farmers to follow is to use integrated pest management practices to closely monitor insect and disease occurrence. Keeping pest and crop management records over time will allow farmers to evaluate the economics and environmental impact of pest control and determine the feasibility of using certain pest management strategies or growing particular crops. Some of the potential adaptation strategies could be developing IPM with more emphasis on biological control and changes in cultural practices, pest forecasting using recent techniques such as simulation modeling and alternate production techniques.

References

1. Andrew, N.R. and Hughes, L. 2005. Diversity and assemblage structure of phytophagous Hemiptera along a latitudinal gradient: predicting the potential impacts of climate change. *Global Ecology and Biogeography*, 14: 249-262.
2. Bale, J.S., Masters, G.J., Hodkinson, I.D., Awmack, C., Bezemer, T.M., Brown, V.K., Butterfield, J., Buse, A., Coulson, J.C., Farrar, J., Good, J.E.G., Harrington, R., Hartley, Z., Jones, T.H., Lindroth, R.L., Press, M.C., Symrnioudis, I., Watt, A.D. and Whittaker, J.B. 2002. Herbivory in global climate change research: direct effects of rising temperature on insect herbivores. *Global Change Biology* 8: 1-16

3. Gupta, H.S. 2011. Climate change and Indian Agriculture : Impacts, mitigation and adaptation. In: *Proceedings of Xth Agricultural Science Congress on Soil, Plant and Animal Health for Enhanced and Sustained Agricultural Productivity*, 10-12th February 2011, ICAR-NBFGR, Lucknow, India, pp.73-81.
4. Hamilton, J.G., Orla, D., Mihai, A., Arthur, R.Z., Alistair, R., May, R.B. and Evan, H.D. 2005. Anthropogenic changes in tropospheric composition increase susceptibility of soybean to insect herbivory. *Environmental Entomology*, 34(2): 479-485.
5. Harrington, R., Fleming, R.A. and Woiwod, I.P. 2001. Climate change impacts on insect management and conservation in temperate regions: can they be predicted?. *Agricultural and Forest Entomology*, 3: 233-240.
6. Lewis, T. 1997. Major crops infested by thrips with main symptoms and predominant injurious species (Appendix II). Thrips as Crop Pests (ed. by T Lewis), pp. 675-709. CAB International, New York, USA.
7. Reiners, S. and Petzoldt, C. 2005. Integrated Crop and Pest Management Guidelines for Commercial Vegetable Production. Cornell Cooperative Extension Publication #124VG
8. Sharma, H.C. 2010. Effect of climate change on IPM in grain legumes. In: *5th International Food Legumes Research Conference (IFLRC V), and the 7th European Conference on Grain Legumes (AEP VII)*, 26-30th April 2010, Anatalaya, Turkey.
9. Williams, R.S., Lincoln, D.E. and Norby, R.J. 2003. Development of gypsy moth larvae feeding on red maple samplings at elevated CO₂ and temperature. *Oecologia*, 137: 114-122.
10. Yamamura, K. and Kiritani, K. 1998. A simple method to estimate the potential increase in the number of generations under global warming in temperate zones. *Applied Entomology and Zoology*, 33: 289-298.
11. Zvereva, E.L. and Kozlov, M.V. 2010. Responses of terrestrial arthropods to air pollution: a metaanalysis. *Environmental Science Pollution Research International*, 17: 297-311.

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